

Lecture 6: Estimating the variance of errors

Economics 326 — Methods of Empirical Research in Economics

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The importance of σ^2

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- The variance of $\hat{\beta}$ depends on the unknown $\sigma^2 = E[U_i^2 | \mathbf{X}]$:

$$\text{Var}(\hat{\beta} | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- If U 's were observable, we could estimate σ^2 by $\frac{1}{n} \sum_{i=1}^n U_i^2$, which is unbiased. This is not possible as U 's are unobservable.
- Using sample residuals instead, $\hat{U}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$, gives a feasible estimator:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2,$$

- $\hat{\sigma}^2$ is biased.

An unbiased estimator of σ^2

- An unbiased estimator of σ^2 is

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2.$$

- Assumptions:
 1. $Y_i = \alpha + \beta X_i + U_i$.
 2. $E[U_i | \mathbf{X}] = 0$ for all i .
 3. $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i .
 4. $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
- Since $\hat{U}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$, dividing by $n-2$ adjusts for estimating two parameters: α and β .

Expressing $\hat{U}_i$ in terms of U_i

- $\hat{U}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$
- $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$, so

$$\hat{U}_i = (Y_i - \bar{Y}) - \hat{\beta}(X_i - \bar{X})$$

- Also:

$$Y_i - \bar{Y} = \beta(X_i - \bar{X}) + U_i - \bar{U},$$

- We have the following relationship between $\hat{U}_i$ and U_i :

$$\hat{U}_i = U_i - \bar{U} - (\hat{\beta} - \beta)(X_i - \bar{X}).$$

- $\hat{U}_i$ is related to U_i , but it is contaminated by the estimation errors.

Expanding $\sum \hat{U}_i^2$

- We have:

$$\hat{U}_i = (Y_i - \bar{Y}) - \hat{\beta}(X_i - \bar{X})$$

- The squared residual is:

$$\begin{aligned} \hat{U}_i^2 &= (U_i - \bar{U})^2 + (\hat{\beta} - \beta)^2 (X_i - \bar{X})^2 \\ &\quad - 2(\hat{\beta} - \beta)(X_i - \bar{X})(U_i - \bar{U}). \end{aligned}$$

- Thus,

$$\begin{aligned} \sum_{i=1}^n \hat{U}_i^2 &= \sum_{i=1}^n (U_i - \bar{U})^2 \\ &\quad + (\hat{\beta} - \beta)^2 \sum_{i=1}^n (X_i - \bar{X})^2 \\ &\quad - 2(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U}). \end{aligned}$$

- To show $E\left[\sum_{i=1}^n \hat{U}_i^2 \mid \mathbf{X}\right] = (n-2)\sigma^2$, we verify the three terms on the RHS:

- **Claim 1:**

$$E\left[\sum_{i=1}^n (U_i - \bar{U})^2 \mid \mathbf{X}\right] = (n-1)\sigma^2.$$

- **Claim 2:**

$$\begin{aligned} E\left[(\hat{\beta} - \beta)^2 \sum_{i=1}^n (X_i - \bar{X})^2 \mid \mathbf{X}\right] \\ = \sigma^2. \end{aligned}$$

- **Claim 3:**

$$\begin{aligned} -2 \cdot E\left[(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U}) \mid \mathbf{X}\right] \\ = -2\sigma^2. \end{aligned}$$

Estimating the variance of $\hat{\beta}$

- Variance of $\hat{\beta}$ conditional on $\mathbf{X}$:

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Estimator of σ^2 :

$$\begin{aligned} s^2 &= \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2 \\ &= \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i)^2. \end{aligned}$$

- Estimator of the variance of $\hat{\beta}$:

$$\widehat{\text{Var}}(\hat{\beta}) = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Standard error of $\hat{\beta}$:

$$\begin{aligned} \text{se}(\hat{\beta}) &= \sqrt{\widehat{\text{Var}}(\hat{\beta})} \\ &= \sqrt{\frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}. \end{aligned}$$

Example in R

- Regress hourly wage on years of education using the `wage1` dataset from Wooldridge:

```
library(wooldridge)
data("wage1")
fit <- lm(wage ~ educ, data = wage1)
summary(fit)
```

Call:

```
lm(formula = wage ~ educ, data = wage1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.3396	-2.1501	-0.9674	1.1921	16.6085

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.90485	0.68497	-1.321	0.187
educ	0.54136	0.05325	10.167	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.378 on 524 degrees of freedom

Multiple R-squared: 0.1648, Adjusted R-squared: 0.1632

F-statistic: 103.4 on 1 and 524 DF, p-value: < 2.2e-16

- Std. Error column: standard errors $\text{se}(\hat{\alpha})$ and $\text{se}(\hat{\beta})$
- Residual standard error: $s = \sqrt{s^2}$; so $s^2 = 3.378^2 \approx 11.41$
- The estimate s^2 can also be extracted directly:

```
summary(fit)$sigma^2
```

```
[1] 11.41352
```

Proof of Claim 1

$$\begin{aligned} \sum_{i=1}^n (U_i - \bar{U})^2 &= \sum_{i=1}^n U_i^2 - \frac{1}{n} \left(\sum_{i=1}^n U_i \right)^2 \\ &= \sum_{i=1}^n U_i^2 \\ &\quad - \frac{1}{n} \left(\sum_{i=1}^n U_i^2 + \sum_{i=1}^n \sum_{j \neq i} U_i U_j \right). \end{aligned}$$

Taking conditional expectations and using the assumptions,

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n (U_i - \bar{U})^2 \mid \mathbf{X} \right] &= n\sigma^2 - \frac{1}{n} \cdot n\sigma^2 \\ &= (n-1)\sigma^2. \end{aligned}$$

Proof of Claim 2

Because $\mathbb{E} [\hat{\beta} \mid \mathbf{X}] = \beta$,

$$\begin{aligned} \mathbb{E} \left[(\hat{\beta} - \beta)^2 \mid \mathbf{X} \right] &= \text{Var} (\hat{\beta} \mid \mathbf{X}) \\ &= \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{E} \left[(\hat{\beta} - \beta)^2 \sum_{i=1}^n (X_i - \bar{X})^2 \mid \mathbf{X} \right] \\ = \sigma^2. \end{aligned}$$

Proof of Claim 3

Note that

$$\sum_{i=1}^n (X_i - \bar{X}) (U_i - \bar{U}) = \sum_{i=1}^n (X_i - \bar{X}) U_i,$$

and

$$\hat{\beta} - \beta = \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Therefore,

$$\begin{aligned} (\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X}) (U_i - \bar{U}) \\ = \frac{(\sum_{i=1}^n (X_i - \bar{X}) U_i)^2}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

Conditionally on $\mathbf{X}$,

$$\begin{aligned} \mathbb{E} \left[(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X}) (U_i - \bar{U}) \mid \mathbf{X} \right] \\ = \frac{\sigma^2 \sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ = \sigma^2. \end{aligned}$$

Putting it together

Using the three expectations above,

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n \hat{U}_i^2 \mid \mathbf{X} \right] &= (n-1)\sigma^2 + \sigma^2 - 2\sigma^2 \\ &= (n-2)\sigma^2, \end{aligned}$$

so s^2 is unbiased for σ^2 .